

TARUN SADARLA

Denton, TX 76201 | 940.765.8671 | tarunsadarla26@gmail.com

tarunsadarla2606.github.io | linkedin.com/in/tarun-sadarla-715026231 | github.com/TarunSadarla2606 | kaggle.com/tarunsadarla

CAREER PROFILE

Medical Imaging AI engineer with an MS in Artificial Intelligence (Biomedical Concentration, GPA 4.00) from the University of North Texas, completed May 2026, and a B.Tech in Computer Science and Engineering from Manipal Institute of Technology. Built and deployed three production clinical AI systems spanning obstetric ultrasound, neuroimaging, and histopathology, each validated against real clinical accuracy thresholds rather than academic benchmarks alone. Comfortable across the full ML lifecycle, from data preprocessing and model development (CNNs, Vision Transformers, U-Nets) through explainability, uncertainty quantification, generative AI integration (LLM-based reporting via the Claude API), and deployment (FastAPI, Docker, HuggingFace Spaces). Particularly interested in oncology and cancer imaging AI, with additional interest across radiology, ultrasound, and pathology applications.

EDUCATION

University of North Texas | Denton, TX

May 2026

Master of Science, Artificial Intelligence, Biomedical Concentration | GPA 4.00

Relevant coursework: AI for Wearables and Healthcare, Applications of AI in Health, Deep Learning, Machine Learning, Advanced Pattern Recognition, Feature Engineering, Fundamentals of AI, Software Development for AI

Manipal Institute of Technology | Manipal, Karnataka, India

Dec 2023

Bachelor of Technology, Computer Science and Engineering | Minor: Data Analytics

PROFESSIONAL EXPERIENCE

Instructional Assistant, Fundamentals of AI (CSCE 5210)

Aug 2025 – May 2026

University of North Texas, Department of Computer Science and Engineering | Supervisor: Prof. Russel Pears

- Supported course instruction across two semesters (Fall 2025 and Spring 2026), covering AI agent architectures, search methods, and optimization techniques.
- Designed and graded assignments and held regular office hours, providing one-on-one debugging support on search strategies and optimization algorithms.
- Maintained consistent positive student feedback across both semesters.

TECHNICAL SKILLS

Medical Imaging and Vision: DICOM, MRI and radiology image analysis, U-Net segmentation, digital pathology, OpenCV, CNNs, Vision Transformers

ML and Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn

Generative AI and LLMs: Claude API integration, prompt engineering, LLM-based clinical report generation

Explainability and Validation: GradCAM++, LIME, MC-Dropout uncertainty estimation, bias and subgroup analysis, Model Cards

Deployment: Docker, FastAPI, Flask, AWS, HuggingFace Spaces, Streamlit

Programming: Python, SQL, MATLAB, C

Big Data and Tools: Hadoop, Hive, PySpark, Cassandra, Git, Anaconda

KEY PROJECTS

TemporalFetalSegNet: Fetal Head Circumference Estimation

Aug 2025 – Dec 2025

- Designed a Temporal Attention U-Net for fetal head segmentation from ultrasound cine-loop video, reaching 97.36% Dice and 1.75mm head circumference mean error, within the ISUOG ± 3 mm clinical threshold.
- Compressed the deployed model by 43.7% using a custom pruning method (Hybrid Crossover Merging) built for this project, with no meaningful loss in accuracy.
- Built a full clinical reporting pipeline, including GradCAM++ attention maps and gestational age estimation via the Hadlock formula, and deployed it live on HuggingFace Spaces and as a standalone web app.
- Co-developed with a research partner under Prof. Xiaohui Yuan.

Autism Spectrum Disorder Detection from Brain MRI

Jul 2023 – Nov 2023

- Developed a CNN and Vision Transformer ensemble for ASD detection from structural MRI; the pure CNN model reached 0.994 AUC across 17 independent acquisition sites (ABIDE-I, 1,067 subjects), and systematically diagnosed ViT underperformance under resource constraints.
- Rebuilt the system with MC-Dropout uncertainty gating, GradCAM++ and LIME explainability, and an LLM-integrated (Claude API) clinical report generator; now deployed live on HuggingFace Spaces.

Histopathologic Cancer Detection System

Aug 2024 – Sep 2024

- Built a 5-layer CNN classifier for cancerous versus non-cancerous pathology image classification, reaching 0.81 ROC-AUC, with a dedicated false-negative subgroup analysis by tissue pattern.
- Deployed as a containerized FastAPI service, framed against EU IVDR Class B regulatory requirements for clinical decision-support software.

Channel Pruning and Compression in CNNs (Hybrid Crossover Merging)

Aug 2025 – Dec 2025

- Designed a structured pruning method that replaces binary filter elimination with regression-based weight synthesis, reaching 2x channel compression on VGG16 with a +0.37% Top-1 accuracy improvement on CelebA.
- Built dynamic safety guards and rollback mechanisms enforcing global and per-class accuracy constraints during optimization; this method was later reused to compress the fetal head model.

WE CARE: Wearable Emergency Cardiac and Fall Response System

Nov 2025

- Built an on-device, real-time AI framework fusing ECG and IMU signals for closed-loop cardiac and fall detection, with no cloud dependency.
- Reached F1 0.99 on arrhythmia detection (MIT-BIH) and F1 0.81 on fall detection (MobiFall), with edge inference latency under 40ms.

OTHER PROJECTS

- Multimodal Sentiment and Emotion Classification (video and audio, MELD dataset, CNN + BiLSTM early fusion).
- Big Data-Driven Customer Segmentation in E-Commerce (AWS EMR/PySpark pipeline, RFM feature engineering, K-Means and Gaussian Mixture clustering).